

Covariate Shift Regularization with Kernels

Suvinava Basak

December 18, 2024

Abstract

Lorem Ipsum

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1 Introduction

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2 Overview: Covariate Shift

Covariate shift occurs when the marginal distribution of the input variables $\mathbb{P}_{\text{train}}(\mathcal{X})$ differs between the training and testing phases, while the conditional distribution of the output given the input $\mathbb{P}(\mathcal{Y}|\mathcal{X})$ remains unchanged.

Mathematically, this is expressed as:

$$\begin{aligned}\mathbb{P}_{\text{train}}(\mathcal{X}) &\neq \mathbb{P}_{\text{test}}(\mathcal{X}) \\ \mathbb{P}_{\text{train}}(\mathcal{Y}|\mathcal{X}) &= \mathbb{P}_{\text{test}}(\mathcal{Y}|\mathcal{X})\end{aligned}$$

The goal is to learn a function f during training, based on data $(X_i, Y_i)_{i=1}^n \sim \mathbb{P}_{\text{train}}(\mathcal{X}, \mathcal{Y})$, that generalizes well on the test distribution $\mathbb{P}_{\text{test}}(\mathcal{X}, \mathcal{Y})$. The change is to adjust for the mismatch between $\mathbb{P}_{\text{train}}(\mathcal{X})$ and $\mathbb{P}_{\text{test}}(\mathcal{X})$, which can degrade model performance.

3 Overview: Kernel Methods

Kernel methods operate by embedding input data $X \in \mathcal{X}$ into a high-dimensional feature space $\mathcal{H}$ through a mapping function $\phi : \mathcal{X} \rightarrow \mathcal{H}$. Instead of explicitly computing the mapping $\phi(X)$, kernel methods compute inner products in the feature space using a **kernel function** $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$, where:

$$k(X, X') = \langle \phi(X), \phi(X') \rangle_{\mathcal{H}}$$

Common kernel functions include:

- **Linear kernel:** $k(X, X') = X^\top X'$
- **Radial Basis Function (RBF) kernel:** $k(X, X') = \exp(-\gamma \|X - X'\|^2)$
- **Polynomial kernel:** $k(X, X') = (X^\top X' + c)^d$

3.1 Covariate Shift in Kernel Methods

In the kernel framework, the objective is to find a function f in the reproducing Kernel Hilbert Space (RKHS) $\mathcal{H}$ induced by the kernel function $k(\cdot, \cdot)$. Given training data $(X_i, Y_i)_{i=1}^n$, we aim to minimize the risk:

$$\mathcal{R}(f) = \mathbb{E}_{\mathbb{P}_{\text{train}}}[\ell(Y, f(X))]$$

where, $\ell(\cdot, \cdot)$ is a loss function (e.g., squared error for regression, cross-entropy for classification). However, under covariate shift the risk we care about is with respect to the test distribution $\mathbb{P}_{\text{test}}$:

$$\mathcal{R}_{\text{test}}(f) = \mathbb{E}_{\mathbb{P}_{\text{test}}}[\ell(Y, f(X))]$$

Since $\mathbb{P}_{\text{test}}(X) \neq \mathbb{P}_{\text{train}}(X)$, the standard empirical risk minimization (ERM) on training data may not perform well on the test data.

Empirical Risk Minimization

We recall the goal of supervised learning: Given some observations $(x_j, y_j) \in \mathcal{X} \cdot \mathcal{Y}$ ($j = 1, 2, \dots, n$) of inputs and outputs (or features/variables) that we call *training data* and given a new $x \in \mathcal{X} = \mathbb{R}^d$, predict $y \in \mathcal{Y} = \mathbb{R}$ (*test data*) with a regression function $f : \mathcal{X} \rightarrow \mathcal{Y}$ such that $f(x) \approx y$.

Given data $(x_j, y_j) \in \mathcal{X} \times \mathcal{Y}$, $j = 1, 2, \dots, n$, the empirical risk is defined by:

$$\hat{\mathcal{R}}(\theta) := \frac{1}{n} \sum_{j=1}^n (y_j - f_\theta(x_j))^2, \theta \in \Theta$$

To obtain a "good" or "reasonable" estimator, we minimize the empirical risk.

3.1.1 Adjusting for Covariate Shift

To account for the shift in the distributions, we introduce **importance weighting**. The idea is to reweight the training data to match the test distribution. If the **Radon-Nikodym derivative** $w(X) = \frac{d\mathbb{P}_{\text{test}}(X)}{d\mathbb{P}_{\text{train}}(X)}$ (i.e., the ratio of the test distribution to the training distribution) is known or can be estimated, we can minimize the weighted empirical risk:

$$\mathcal{R}_{\text{weighted}}(f) := \frac{1}{n} \sum_{i=1}^n w(X_i) \ell(Y_i, f(X_i))$$

This ensures that the training data is effectively "corrected" to resemble the test distribution.

3.1.2 Kernel Ridge Regression under Covariate Shift

Let us consider **Kernel Ridge Regression (KRR)**, a kernel-based method for regression that minimizes a regularized empirical loss (with the *least square loss*). Without covariate shift, the solution $f \in \mathcal{H}$ is found by solving:

$$\hat{f} = \arg \min_{f \in \mathcal{H}} \left\{ \frac{1}{n} \sum_{i=1}^n (Y_i - f(X_i))^2 + \lambda \|f\|_{\mathcal{H}}^2 \right\}, \quad (3.1)$$

where $\lambda > 0$ is a regularization parameter that controls overfitting.

To adjust for the covariate shift, we apply the importance weights to the loss function:

$$\hat{f} = \arg \min_{f \in \mathcal{H}} \left\{ \frac{1}{n} \sum_{i=1}^n w(X_i) (Y_i - f(X_i))^2 + \lambda \|f\|_{\mathcal{H}}^2 \right\} \quad (3.2)$$

The solution to this problem can be written in terms of the kernel matrix $K \in \mathbb{R}^{n \times n}$ with entries $K_{ij} = k(X_i, X_j)$, and the importance weights can be incorporated into a diagonal

matrix $W = \text{diag}(w(X_1), \dots, w(X_n))$. The closed-form solution for the KRR estimator under covariate shift is given by:

$$\hat{f} = (KWK + \lambda I)^{-1}KWY, \quad (3.3)$$

where $Y = (Y_1, \dots, Y_n)^\top$ is the vector of training labels. This solution adjusts for the shift by weighting the kernel matrix and labels according to the test distribution.

3.1.3 Estimating Importance Weights

In practice, the true density ratio $w(X) = \frac{\mathbb{P}_{\text{test}}(X)}{\mathbb{P}_{\text{train}}(X)}$ is often unknown. There are several strategies to estimate $w(X)$, including:

- **Kernel Density Estimation (KDE)**: Estimate both $\mathbb{P}_{\text{train}}(X)$ and $\mathbb{P}_{\text{test}}(X)$ using kernel density estimators and compute the ratio.
- **Logistic Regression-based Density Ratio Estimation**: Train a logistic regression model to distinguish between samples from the training and test distributions. The estimated probabilities can be used to approximate the density ratio.
- **KLIEP (Kullback-Leibler Importance Estimation Procedure)**: Directly estimate the ratio by minimizing the Kullback-Leibler divergence between the reweighted training distribution and the test distribution.

Once $w(X)$ is estimated, it can be plugged into the weighted KRR formulation to adjust for the covariate shift.

3.1.4 Self-Attention for Kernel Methods under Covariate Shift

Another approach is the **self-attention mechanism**, where the importance weights $w(X)$ are learned dynamically along with the model parameters. This can be formulated as a joint optimization problem, where the objective is to **minimize the empirical loss with respect to both the regression function f and the weights $w(X)$** . The optimization alternates between updating the model parameters f and the weights $w(X)$ using gradient-based methods.

In kernel-based methods, this can be implemented by introducing learnable parameters w_i for each sample, and optimizing both the KRR objective and the weights simultaneously:

$$\min_{f, w} \left\{ \frac{1}{n} \sum_{i=1}^n w(X_i) \ell(Y_i, f(X_i)) + \lambda \|f\|_{\mathcal{H}}^2 \right\} \quad (3.4)$$

subject to constraint on $w(X)$ (e.g., positivity, normalization). This allows the model to adapt the importance weights based on the data characteristics, potentially improving performance under complex covariate shift scenarios.

Bibliography